DiamondTrading: Transparent-Queue Turbo Markets

A Public FIFO Payout Queue with Claim-Level Transparency, Manipulation-Cost Position Caps, and Counter-Cyclical Anchored Loss-Mint Tokenomics for 1000x Leveraged Wagering

DiamondTrading Research

Version 0.3 · October 2026 · supersedes v0.2 (queue-bounded); implemented and fuzz-verified

Abstract

Hyper-short-dated leveraged wagering — post margin, drag leverage to 1000x, bust or win within seconds — reached production scale and then collapsed under its own accounting: the deployed lineage descending from the Martin Galer queue/emission design accumulated dozens of audit findings, including outcome-manipulation and mint-accounting exploits [1][2]. v0.1 of this paper eliminated the queue with full reserves (cannot carry 1000x menus); v0.2 re-admitted the queue but bounded it with brakes that turned out to misprice the game itself. We present DiamondTrading v0.3, the papertrade-lineage design taken to its honest end: deferred payment exists, is public to the wei, and is never used as an excuse to stop the machine. Layer one is a transparent FIFO payout engine: profit the pool cannot cover becomes permanent, order-preserving on-chain claims; loser money feeds the pool directly; every settlement inline-pays the oldest claims; there is no queue ceiling, no side bucket, no pause switch — the product never stops selling, and every claimant reads their exact position and owed amount from the contract itself (§4). Layer two is the measurement-driven quoting engine: deepest-venue routing, VWAP-of-real-fills settlement anchors, manipulation-cost caps κλω/ω̃ with no treasury terms (§6). Layer three is the counter-cyclical anchored mint: DIAMOND emission starts at 200% of the signed realized-net-revenue anchor, tightens −250bps/epoch as revenue grows (sustained success siết emission), and re-eases +1000bps/epoch when revenue weakens — capped at 200%, floored at 30%, absolutely epoch-capped (§7). The system is implemented — both payout engines, verifiers, tokenomics, and the off-chain engine — with 130 contract assertions, 51 engine assertions, 43 security regressions, and five fuzz seeds mirroring every money lane after every operation (§11). We state the unbounded-debt risk, the attester-quorum trust assumption, and the licensing constraint frankly (§12).

Provenance convention. Every externally checkable claim carries exactly one tag, declared in its reference entry (§13): [CITED] measured/sourced, [COMPUTED] derived from cited data or the shipped code, [RECALLED] author's recall of public market structure, [ASSUMED] initial design parameter (governance-tunable).

1 Introduction

Papertrade demonstrated that the retail gamblers' terminal is not a spot market with leverage controls; it is a turbo market: margin in, leverage to 1000x, bust price displayed, outcome in under a minute, plus a token kicker minted on losses that made losing feel like progress. It also demonstrated the failure mode of building that product on hidden accounting and gross-loss emissions: the accountability debt surfaced as 42 audit findings, two of them critical [2]. v0.1 went to full reserves and could not sell the 1000x menus the product exists for; v0.2 bounded the queue with a QMAX brake and a loser-custody rule, and testing showed both were theatre: a brake that stops sales when the house is weakest is a death sentence, and custody rules that exclude loser money from the pool starve the very queue they protect. v0.3 deletes them. What remains is the papertrade design told honestly: a FIFO queue that never blocks sales, loser money that feeds the pool directly, stall refunds that never gate, and claim-level transparency that tells every user exactly where in line they stand. Deferred payment is declared, publicized per-claim — never patched silently.

The remaining text is organized as follows. §2 recalls the lineage and the audit taxonomy we design against. §3 defines actors, assets, and the trust model. §4 gives the transparent-FIFO payout engine and its lane invariant. §5 defines the instruments' payoffs. §6 specifies the GALE off-chain engine (routing, settlement persistence, caps, breaker, pricer). §7 specifies the smart mint. §8 treats verifiers. §9 runs the security analysis class by class. §10 covers economics. §11 reports implementation and test evidence. §12 lists limitations honestly; §13 is the reference and provenance ledger.

2 Lineage and the audit taxonomy

Martin Galer (2024-era) introduced the zero-LP bootstrap: queue winners, pay them from later flow, and emit a token on losses advertised as +EV [1]. Its solvency-ratio invariant was elegant; its deployment was not. The empirical record: queue and side-bucket accounting exploits (H-01..M-05), BBO-midpoint outcome manipulation with no deviation or persistence checks (KV-01), straddle-style short optionality handed to attackers (KV-02), and governance degradable to 1-of-2 signers (L-04) [2]. Our benchmark in §9 is this taxonomy; a modern turbo market fails by repeating any of it.

Design stance. We copy papertrade's toy (margin, leverage, bust price, seconds, mint on loss) and keep its one good instinct — the game never stops — while straightening every ledger it left crooked: the queue is FIFO and public (each claim addressable; position readable on-chain), loser money flows into the pool the instant it is realized, settlement is VWAP-of-fills not midpoints, and the mint is anchored to signed real revenue with a rate that reacts to the protocol's own health [2][3][4].

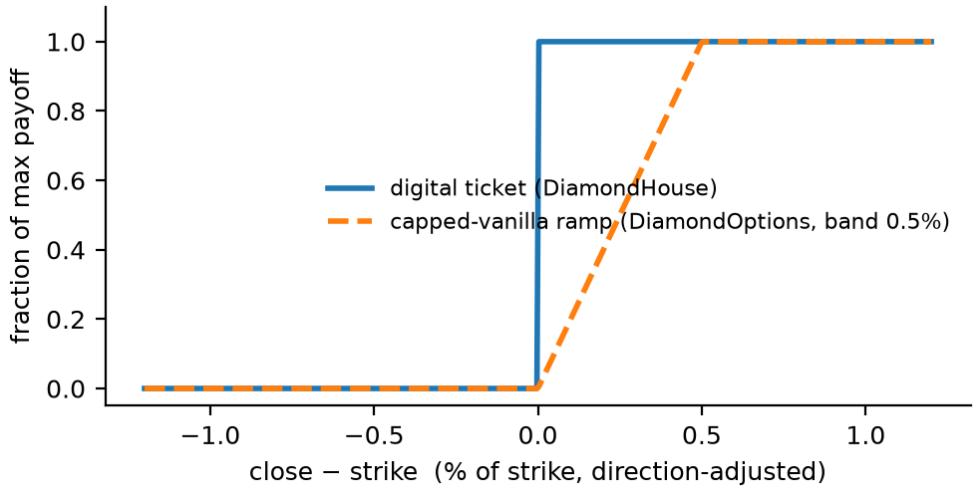


Figure 2 — The two issuance shapes: a digital ticket (pay M·S or 0) and a capped-vanilla ramp (pay intrinsic up to the band). Neither escrows the maximum at sale; unpaid winning claims enter the public FIFO queue.

3 Actors, assets, trust

actor	role	what they can and cannot do [COMPUTED: contract code 5]
player	posts margin, receives tickets	stake back instantly on any correct claim; profit paid now or via FIFO queue; claimInfo says exactly where they stand; 24h stall ⇒ permissionless stake refund; nothing can pause sales
LP	pool capital backing payouts	first-loss by construction (pool pays before any queue forms); residual pool NAV after the exit lock; no separate LP revenue floor [5]
quoter	off-chain pricing engine	signs one-use quotes (σ-priced, κλω-capped, ≤30s quote validity + signed position TTL); cannot settle or move funds; worst-case = bad menus bounded by on-chain class clamps
attester set (2-of-3)	price persistence verification	cannot settle alone; cannot forge (2 keys); stall ⇒ permissionless stake refund; threshold floored at 2 on-chain
budgetAttester	revenue anchor for the mint budget	can only lower emissions; cannot touch principal; bonded, rotating
owner (governance)	config + per-class (enabled, maxMult, profitCap)	global envelope 1.05x–1000x fixed on-chain; production path = Safe 2-of-3 + 7-day timelock [5]

Underlyers at launch [3][ASSUMED: menu], per-class enabled/cap/mult governance on-chain: BTC & ETH — 1000x class (Binance-routed: $12.0–13.1B resp. $8.9–9.9B 24h); SOL & HYPE — 500x class (SOL Binance-settled at $1.3–1.5B; HYPE Hyperliquid-native, $327–392M on HL vs $399M Binance); BNB — 200x class (Binance $280–366M); S&P500; & NASDAQ-100 equity perps — 100x class (Hyperliquid HIP-3 only, ~$310–332M [4]). Routing per §6.1 [5].

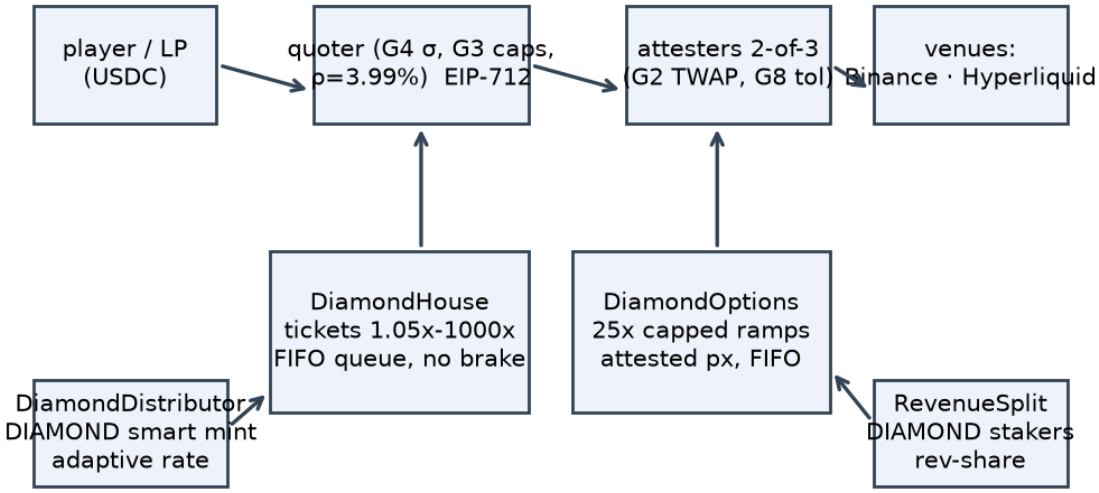


Figure 1 — System map: players post USDC; the quoter prices quotes; attesters verify venue prints; the engines pay now or queue FIFO (publicly); the mint and the rev-split close the loop.

4 The transparent FIFO payout engine

Let B be the contract's USDC balance. Three accounted lanes partition obligations: pool (house money: fed by loser stakes, premiums, and LP deposits), open (sum of live stakes, fully player-owned), and credits (payouts refused by the token, withdrawable later). Donations are unallocated surplus. The liability side is debt — the sum of queued, unpaid profits. Debt is NOT cash: it is the house's public obligation, tracked to the wei. At every block:

$$B \geq pool + open + credits, \quad debt = \Sigma\ unpaid\ queued\ entries \geq 0$$

Each lane is a first-class storage variable, never derived; lanesIntact() re-verifies the solvency inequality after every settlement and every treasury send [5]. v0.3 has NO hidden accounting surfaces: v0.2's side bucket (a second custody lane that answered “where do loser funds sit while the queue waits”) is deleted — loser money lands in the pool, and the pool pays the line. If the queue outruns the pool, every waiting claimant can see it happen.

Sale. A ticket of stake S and maximal profit P_{max} sells iff its class is enabled and P_{max} ≤ the class profit cap. There are no other gates — v0.2's QMAX (a sales brake that engaged precisely when the house was under stress) is removed by explicit directive: stopping sales in a drawdown starves the payout stream, i.e. guarantees the worst outcome. Selling through the storm is the recovery mechanism; transparency is the safety mechanism. There is also no pause switch: neither sales, settlement, refunds, nor LP exit can be frozen by anyone — ownership tunes menu config only [5].

Settlement of a correct claim returns the stake in full, then pays min(pool, net) where net = (1−ρ)·profit, ρ=3.99%. The unpaid remainder appends a FIFO queue entry; entry ids are a zero-folded array with a windowed (head, live) view [5]. While debt > 0, correct claims enqueue in full — paying a newcomer ahead of an older claimant is unrepresentable. Lost stakes (KNOCKED) and the forfeited 85% of EXPIRED tickets flow into pool unconditionally (papertrade rule); every settlement then inline-pays up to INLINE_HARVEST=4 queue entries from pool (≈1,950 gas marginal per entry [COMPUTED]); any caller may harvest(1..64) for more, and LP exit NAV slices are computed on the same lane arithmetic.

Claim-level transparency (the frontend promise). queueOf maps every ticket to its queue entry; claimInfo(ticketId) returns (hasDebt, amountOwed, position, totalDebt) — position 1 “you are next”. The UI rule is strict: when the queue is empty, nothing is said; when a user holds a queued claim, the app states their position and amount (`mày đang đứng thứ X, được nợ Y USDC`). Debt is a broadcast fact of the system, not a warning light for an ad-hoc brake: users

price it in, LPs see a measurable liability, and opportunistic mints (§7) react to it.

Theorems (fairness and liveness). (a) No cut-in-line: queue payments issue only to entry ids ≥ queueHead in ascending order, and queueHead advances only when the head is fully paid (property G4, asserted on every fuzz operation). (b) Claims never die: no code path expires, cancels, or re-keys a queued amount — a winner's piece of the pie exists until paid in full. (c) Nothing freezes: sales, settlement, stall-refund (24h), harvest, and LP-request/withdraw are always live; there is no governance pause to yield, and the owner can never gate user funds [5]. (d) Recovery is the steady state, not a fallback: in a debt phase, every loser dollar and every new LP dollar pays the line FIFO; in a clean phase the same dollars feed the waterfall (§10). □

Winning-trade fee split. ρ=3.99%: 0.99% of House profit (or Options gross payout) streams to founderWallet, and the remaining 3.00% streams to RevenueSplit for DIAMOND stakers. The owner sets the founder address later on both engines; while it is zero, that slice folds into the active staker sink. Rounding dust goes to stakers [5].

Honest accounting and cash gate. The full ρ haircut is recorded in Settled/OSettled for accounting; the epoch anchor uses stake−payout so fees count exactly once. Actual cash streaming is narrower: a queued or partially paid claim streams zero; a fully paid claim streams only if the post-payout pool covers the outgoing amount. Otherwise the unstreamable amount stays pooled. With no active stakers, their slice also stays pooled instead of becoming stranded in RevenueSplit. FeeStreamed records committed amounts; token-refused transfers become credits. The Options engine books premium only as house income; the unpaid slice of maxPay is a queue-bound liability haircut, not revenue [5].

5 Instruments

A position posts S, picks multiplier M ∈ [1.05x, class max] and direction d ∈ {LONG, SHORT}; the on-chain envelope spans 1.05x–1000x, capped per class (BTC/ETH 1000x, SOL/HYPE 500x, BNB 200x, equity perps 100x) [5]. Entry K is the routed venue's trade VWAP over ω=2s at buy; the bust price is K·(1∓1/M), displayed pre-trade [ASSUMED: ω]. Three attested outcomes at expiry τ ∈ [30s, 300s]: WON (correct direction) returns stake + profit(M,S,close) with ρ shaved off profit; the cash schedule of §4 applies (pool now, rest FIFO). KNOCKED (wrong direction): stake forfeits to pool. EXPIRED (boundary): 15% stake rebate, rest forfeits; settlement is refused before expiry so band-β pool readings are always post-close [5]. Two shapes share the schedule:

shape	payoff at close C	contract
digital ticket	stake S back + (1−ρ)·(M−1)·S when correct, else forfeit	DiamondHouse
capped-vanilla ramp	(1−ρ)·maxPay · min(|C−K|, band·K)/(band·K); premium is house income	DiamondOptions

The ramp is the options-flavoured mode: band = strike-relative cap, premium-funded, maxPay class-capped; house income is the premium only (§4 note). Graded wins keep engagement at sane multipliers; the digital shape is the papertrade-exact bust loop. Neither shape escrows winnings at sale — that is exactly the bootstrap property — and both bind to the same FIFO machinery.

6 The GALE engine

6.1 Venue routing (G1)

Asset a routes to the venue maximizing measurable depth within the ticket band, re-scored at 5s cadence across Binance (spot+futures) and Hyperliquid (WSS; L1Read proofs later) [ASSUMED: cadence]. Measured 24h quote volumes (Binance + HL, Oct 2026 [3][4]) drove launch routing: BTC $12.0–13.1B vs HL $2.3–3.0B and ETH $8.9–9.9B vs HL $0.7–0.9B → majors single-read on Binance; SOL $1.3–1.5B vs HL $125–152M → Binance-settled; BNB

$280–366M → Binance; HYPE ~$0.40B Binance vs $327–392M HL → Hyperliquid stays its native home; S&P500; / NASDAQ-100 (HIP-3) exist only on HL at $310–332M → HL-only with κ=1/4 [3][4]. Additional venues re-enter only after runtime measurement: the launch router is two-venue by choice, not by default [3].

6.2 Settlement persistence (G2)

Never midpoints. Both anchors are volume-weighted means of real fills over their ω windows: VWAP(t) = Σ $p_i q_i$ / Σ q_i over [t−ω, t]; an empty window yields NO-PRINT, an honest failure that routes to re-attestation or stake refund, never to a guessed price [COMPUTED: engine spec 6]. Forging a persistent print costs real executed volume through the whole window — the KV-01 class (midpoint twitch) is gone by construction [2].

6.3 Manipulation-cost position caps (G3)

The attack that matters: dominate the settlement reference over its measurement window. Its cost is lower-bounded by organic flow $\lambda_a \cdot \omega$ with λ_a the routed venue's measured 24h quote volume per second (EMA-smoothed, floored at its 10th percentile) [ASSUMED: estimator]. The per-position maximal payout obeys:

$$P_{max}(a,\omega) = min(\ \kappa \cdot \lambda_a \cdot \omega/\tilde{\omega},\ \ classCap_a\ ), \quad \tilde{\omega} = 10s\ reference$$

with κ = 1/4 single-read and 1/3 dual-read (an attacker must move both venues to pass G8) [ASSUMED: κ]. Note what is absent: no treasury term. v0.1's T/60 and 0.3T treasuries measured what the vault could afford; under transparent-FIFO bootstrap the pool may legitimately be zero, so the honest ceiling is what the market can cost an attacker — plus the static on-chain class cap as outer wall (G3 only tightens). The ceiling collapses automatically in thin sessions: off-hours equity perps shrink toward zero without human scheduling. Figure 2 plots launch ceilings against measured λ [3][4].

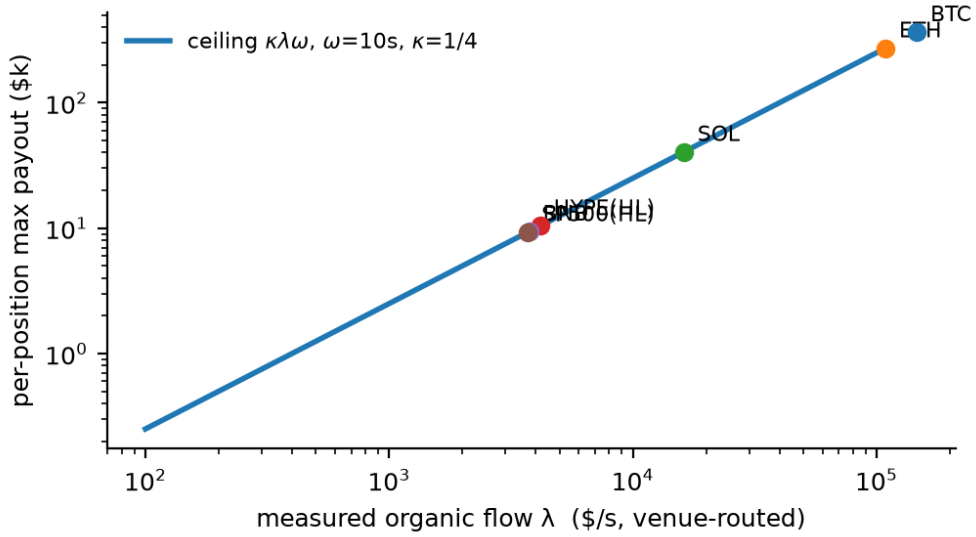


Figure 3 — Per-position profit ceiling κλω (ω=10s) vs measured organic flow, launch assets annotated (log-log; Oct-23 λ). On-chain class caps are the static wall; G3 can only tighten.

6.4 Vol-spike breaker (G4)

If realized σ over 30s exceeds 4× the 30-day median, the quoter stops signing for 60s on that asset [ASSUMED]. In-flight quotes die by their ≤30s TTL on-chain [5]. Stale-quote sniping around macro prints — the standard ticket-shop harvest — has no window.

6.5 Pricer: fair multiplier from measured σ, shaved by ρ

For a zero-drift price, first passage to the ∓1/L barrier within τ is approximately, by the reflection principle:

$$P_{bust}(L,\sigma,\tau) \approx 2\cdot\Phi(\,-(1/L)\,/\,(\sigma\sqrt{\tau})\,),\quad M_{fair} = 1/(1-P_{bust}),\quad M_{offer} = M_{fair}\cdot(1-\rho)$$

with ρ = 3.99% the published winning-claim haircut [ASSUMED: pricing rule]. Offers clamp to the on-chain envelope (1.05x–1000x, per-class tighter) and refuse when the fair multiplier is below the floor — the engine does not sell products it cannot price honestly [5][6]. Figure 4 shows the bust surface.

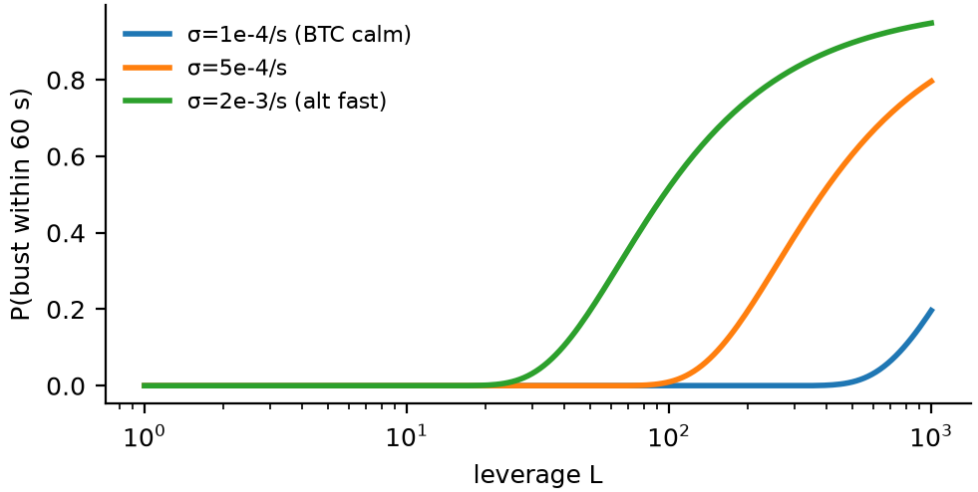


Figure 4 — Bust probability vs leverage for three volatility regimes (τ=60s; reflection-principle model). The menu lives where the curves are informative; extreme L converges to coin-flip and is priced as such.

The 3.99% sticker is below the recalled competitor range of 4–6% [RECALLED: market range]. This is a fee-rate comparison, not a claim about every venue's all-in execution cost; G2+G4 address adverse selection and G3 bounds the tail per position [COMPUTED: from the design above].

7 Smart-mint v3: counter-cyclical emissions losers cannot farm

Loss-minting is the retention engine and was the original protocol's exploit surface. The v3 mint keeps the emotion (losing earns), removes the farm, and replaces any fixed calendar with the rate curve a launch actually needs — rich at boot, siết as the project proves itself, nới again the moment it stumbles:

defense	rule	consequence
net-loss epochs	weight base = max(0, Σstake − Σpayout) per address per 6h epoch	win-then-lose cycles cannot manufacture mintable losses
fee-squared weight	w = √(netLoss · feesPaid) × loyalty (1.25 if staked DIAMOND)	mint requires real rake; √ forces 4× capital for 2× weight
honest anchor	anchor = max(0, Σstake − Σfee-net payout), signed by budgetAttester (USDC 1e6)	budget counts realized net revenue once, not projections; the 200% launch rate is an explicit subsidy, not a below-cost farming guarantee [5]
launch bait	rate boots at 200% of anchor, absolute epoch cap on top	early users are paid like early users
growth siết	anchor > prevAnchor ⇒ rate −250bps/epoch, floor 30%	sustained revenue tightens emission automatically — success, not a vote, is what siết
weakness nới	anchor < prevAnchor ⇒ rate +1000bps/epoch, ceiling 200%	shrinking revenue re-opens the tap, fast, to win flow back [COMPUTED]
hard ceiling	mint ≤ epochCap per epoch (default 2M DIAMOND, ≤5M ceiling, lowerable, never raisable)	sybils split a fixed pie; pro-rata yields no scale [COMPUTED]

The merkle distributor enforces the adaptive schedule on-chain against the signed anchor — EIP-191 over (chainId, distributor, epochId, anchor, priceX1e8), with aggregate epoch claims capped on-chain. Price p(e) has no on-chain reference until DIAMOND transfers open, so the attester signs it (bootstrap p(e) ≈ $0.01; TWAP-chained after). Anchors signed under a wrong domain are rejected by the on-chain identity probe [5]. DiamondToken itself boots non-transferable (allowlist-only protocol moves); public transfer is a one-way flag flipped after launch liquidity criteria, never re-lockable [5]. Weights come from the engine ledger [6]; root/post/claim is covered end to end (§11). DIAMOND carries cashflow via RevenueSplit (pull-only notifyRevenue; seven-day streams with cumulative-index snapshots), which is what makes an asset handed out on losses worth holding rather than dumping. Its cashflow now includes the direct 3.00% winning-claim stream (plus the founder slice while that address is unset), subject to §4's cash and active-staking gates.

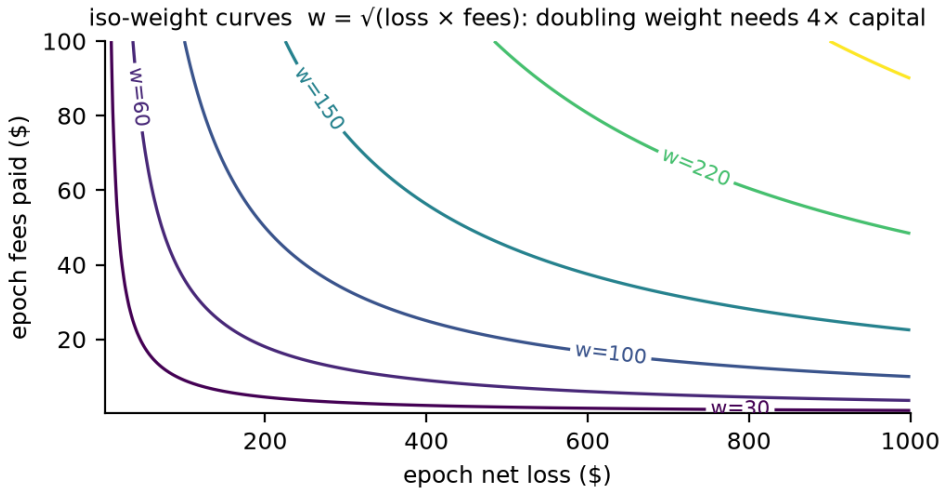


Figure 5 — Iso-weight curves of w=√(loss·fees): weight grows sub-linearly in both inputs, so scale buys nothing and only sustained, rake-paying participation accumulates emission share.

8 Verifiers and the endgame

Settlement inputs are M-of-N EIP-712 attestations (default 2-of-3; the threshold setter is hard-floored at 2 on-chain — L-04 cannot recur) [5]. Attesters each read the routed venue's trade tape independently; dual-listed assets require the two venues' prints within tolerance before anyone signs [ASSUMED: tolerance][COMPUTED: engine]. Anyone may relay a passing proof; no attester can settle alone, block settlement, or fake a stall. If all attesters stall for 24h, every live position unwinds to its stake permissionlessly [5]. Can the code just READ the price itself? Honest no, for launch. (i) Binance-settled assets (BTC/ETH/SOL/BNB) have no on-chain price oracle on HyperEVM — there is nothing canonical for a contract to read. (ii) Even for HL-native assets, L1Read precompiles expose marks/spot prices at this instant; settlement needs the VWAP of real fills over the ω window, a tape query the precompiles do not answer — a single mark read is exactly the BBO-twitch surface KV-01 warned about [2]. The viable endgame is a TWO-STEP design: attesters sign the window VWAP and an L1Read sanity circuit rejects any attested close deviating from the mark by >x bps — keeping the quorum while bounding its worst case [5][ASSUMED: x].

9 Security analysis

Method: replay the audit taxonomy of record [2] and ask for each class whether it exists here.

exploit class (Guardian id)	their habitat	status here	mechanism

payout queue accounting / hidden debt (H-01..M-05)	queue vs ledger divergence	contained	donation-safe cash lanes + pull credits; lanesIntact() solvency; debt is a public per-claim read (claimInfo)
BBO-midpoint settlement (KV-01)	single-source midpoint	absent	VWAP persistence G2 + 2-of-3 tolerance; L1Read cross-check is the stated endgame
short optionality / straddle (KV-02)	unpriced option grants	bounded	payout shapes closed-form; class clamps on-chain; no free bands
unconstrained bet sizing (M-02/05)	flat caps vs flow	closed	G3 κλω/ω̃ dynamic caps + on-chain class profit caps (§6.3)
gross-mint / mint accounting (KV-04)	loss-mint farming	bounded	net-epoch √-weights + domain-bound aggregate budget + absolute cap; 200% launch subsidy is explicit (§7)
keeper/governance capture (L-04, I-20)	1-of-2, timelock→0, pause-gating	bounded	attester threshold floor 2; no global pause; owner still rotates critical roles (§12)
stale settlement / trapped funds (L-11)	keeper stall	closed	24h permissionless refund; settlement/harvest always live
quote replay / intent abuse (KV-14)	WESig re-use	closed	signed position TTL + one-use digest + buyer/domain binding; original and malleable replays rejected
sybil multi-account hedge-mint (new)	long+short pairs across wallets	bounded, not necessarily below cost	anchor counts stake−fee-net payout once; aggregate budget ≤ rate×anchor; launch subsidy can make farming profitable by design (§7)

2026-10-24 remediation. Donation-safe lanes, one-use signed-TTL quotes, seven-day reward streams, aggregate/domain-bound epoch budgets, exact sink allowances, pull credits, zero-entry rejection, and persistent orphan-pool virtual shares with true-empty reset are covered by 43 dedicated regressions [8]. The proposed pool−queueDebt LP NAV patch was not applied: queueDebt remains deferred debt, not a senior LP reserve; the report PoC withdrew only the later LP's own deposit.

Residual risks, stated: attester liveness/collusion above floor (mitigated, not eliminated); κλω is a bound, not a proof of manipulation cost (books can thin faster than flow); equity-perp single-venue windows degrade G8 to single-read; the quoter's strike/σ inputs are trusted within clamp bounds [12]. Each has a named on-chain failsafe; none is silent.

10 Economics

Revenue waterfall on realized edge, gated by queue health [5]: distributable = (pool − TARGET_POOL) × 80%, callable only when the queue is fully clean (debt = 0 — during a payout crisis 100% of flow works the line, none leaks to distributions). The shipped House sends the distribution to RevenueSplit; there is no separate code-enforced claimLP or ≥60% LP floor. LP returns instead remain in residual pool NAV, while notified revenue streams to DIAMOND stakers. LPs underwrite variance and are first-loss: the pool pays winners before any queue forms, and exits settle pro-rata against actual pool cash, not pool−queueDebt [5]. The growth loop: adaptive mint pays launch-rich rebates while flow proves itself (§7); staker rev-share keeps the losing-but-loyal cohort invested; ρ=3.99% plus graded ramp wins drive flow; flow deepens LP returns, shrinks observed debt phases, and the λ-derived caps permit larger honest tickets.

Fee stance versus incumbents: one headline winning-claim rate, ρ = 3.99%, published, split 0.99% founder / 3.00% DIAMOND stakers, with no deposit/withdrawal/turnover fee [ASSUMED]. It sits just below the recalled 4–6% competitor range [RECALLED]; this is arithmetic about the sticker rate, not an unsupported all-in-cost comparison. The founder address is initially unset and owner-configurable later; until then active stakers receive the full

3.99% when streaming is cash-covered.

11 Implementation status and evidence

component	artifact	evidence
payout engines	DiamondHouse v3, DiamondOptions v3 (transparent FIFO) [5]	130/130 contract assertions green: FIFO ordering, exact-debt partials, inline harvest from pool, no-brake sales under arbitrary debt, absence of any pause switch, stall refunds, EXPIRED rebate timing, claimInfo position/amount reads, exact 0.99/3.00 fee split, founder-unset fallback, no-staker and net-only pool boundaries, lane solvency B ≥ pool+open+credits
verifiers	MultiAttesterVerifier, MultiPriceVerifier [5]	2-of-3, floor-2, tolerance paths, wrong-domain signature probes
tokenomics	DiamondDistributor (adaptive rate + domain-bound anchor), RevenueSplit (seven-day streams), DiamondToken [5]	launch 200%, adaptive clamps, aggregate epoch claims, wrong-domain/unset-attester rejection, double-claim block, JIT/multi-staker streaming, transfer guard + one-way enable tested
engine	engine/ (venues, caps, pricer, quoter, attester, ledger) [6]	51/51 assertions incl. ganache interop: engine-signed quotes sell, engine attestations settle, engine merkle posts/claims on-chain; caps carry no treasury terms; adaptive-rate mirror and Options fee-stream split verified
security + fuzzing	test/security.test.js + 150-op seeds 1337/7/42/90210/555 [8]	43/43 security regressions and all five fuzz seeds green: donations, TTL/replay/malleability, JIT/multi-staker fees, sink allowances, epoch/domain replay, orphan/whale/dust LP attacks and true-empty recap, blocklisted payees, zero entries, deploy guards; lane/fee mirrors green

12 Limitations (the honest page)

(i) Deferred payment is a real risk by construction, and v0.3 removes the last artificial brake. There is no QMAX: debt can exceed any bound, sales never stop, and debt is extinguished only by loser flow and LP backfill paying the line FIFO. What users hold against this is information, not promises: each claim is permanent, order-preserving, exactly priced, and individually readable; totalDebt is a public broadcast; distributable() hard-zeros during any debt phase; and 24h stalls refund in full. Timing is a market variable — declared, not laundered into a brake. (ii) Settlement is attester-quorum, not cryptographic trade-print proof; the L1Read verifier is the largest open item. (iii) The collared σ inputs come from the quoter within clamp bounds; dual-print anchored strikes are specified, not shipped. (iv) κλω is a lower bound assembled from estimator assumptions; stress may invalidate the estimator faster than the breaker reacts. (v) Equity perps trade 24/7 on one venue; off-hours is handled by cap decay plus a quoter curfew, both [ASSUMED]. (vi) DIAMOND's bootstrap is non-transferable by design until a one-way flag; hold-horizon risk is declared. (vii) Regulatory: an offshore, unlicensed distribution of this product scores zero on a three-year horizon; licensing (or licensing-grade geofencing) is a launch gate, not a growth item [ASSUMED]. (viii) Fee cashflow is conditional, not a guaranteed per-win transfer: queued/partial claims and insufficient post-payout cash do

not stream, and the staker slice stays pooled when nobody stakes. The founder wallet is a later owner configuration; its zero-address fallback favors active stakers. (ix) queueDebt is not a senior LP reserve, so LP exits and unpaid winners can coexist; pre-LP orphan assets are intentionally locked in virtual shares. (x) Governance remains powerful: no new owner timelock is shipped in this patch. (xi) Tests use mock/blocklist-capable tokens on ganache, not real USDC or a live deployment; no formal verification or independent audit sign-off is claimed.

13 References and provenance ledger

[1] [CITED: Martin Galer whitepaper, docs.papertrade.xyz, fetched Oct 2026] queue-based payout bootstrap, solvency-ratio invariant, loss-emissions.

[2] [CITED: Guardian Security Review of Papertrade Protocol, final report Oct 10 2026] 42 findings; KV/H/M/L ids as used throughout this paper.

[3] [CITED: api.binance.vision snapshots fetched 2026-10-22 + user-provided Binance 24h reads 2026-10-23; Bybit evaluated + dropped (geo-blocked)] spot depths/24h volumes in research/LIQUIDITY.md.

[4] [CITED: user-provided Hyperliquid screenshots, app.hyperliquid.xyz, 2026-10-20] HYPE OI/volume, market list incl. HIP-3 equity perps.

[5] [COMPUTED: diamondtrading/contracts/ DiamondHouse.sol v3, DiamondOptions.sol v3, Verifiers.sol, DiamondDistributor.sol, DiamondToken.sol, RevenueSplit.sol — this repository] all on-chain behavior cited above.

[6] [COMPUTED: diamondtrading/engine/ + test/engine.test.js + test/fuzz.js — this repository] engine behavior cited above.

[7+] [ASSUMED]/[RECALLED] parameters (κ, ω, ρ, tolerances, splits, curfews, market fee ranges) as marked inline.

[8] [COMPUTED: diamondtrading/SECURITY.md + test/security.test.js + test/fuzz.js — this repository] 2026-10-24 finding disposition and dedicated regression evidence.

TÓM TẮT — bản tiếng Việt

DiamondTrading v0.3 là sàn turbo kiểu papertrade (nạp margin, đòn bẩy đến 1000x ở BTC/ETH, biết giá bust trước khi bấm, thắng thua trong 30–300 giây, thua được mint DIAMOND) được dựng lại từ nền: (1) tiền thắng trả ngay từ pool nếu đủ; phần chưa đủ vào hàng đợi FIFO — không ai được vượt hàng, không có trần nợ QMAX, sàn không bao giờ ngừng bán; mỗi claim tự đọc được vị trí trong hàng đợi và số tiền đang được nợ qua claimInfo (app chỉ thông báo khi thực sự có nợ); tiền thua đổ thẳng vào pool — không side bucket, không lệnh sweep; không có nút pause nào, không ai đóng băng được tiền của ai; (2) giá settle là VWAP của lệnh khớp thật trên sàn sâu nhất từng tài sản (BTC/ETH/SOL/BNB đi Binance; HYPE đi Hyperliquid; S&P500;/NASDAQ100 chỉ có trên HL HIP-3); trần lệnh = min(κλω, cap lớp on-chain); (3) mint khi thua nhưng chống farm: lỗ ròng theo epoch, trọng số √(lỗ × phí) × hệ số trung thành; ngân sách = rate × doanh thu thật có chữ ký, với rate THÍCH ỨNG: khởi động 200% (giai đoạn dụ anh em), doanh thu tăng thì siết dần −250bps/epoch về sàn 30%, doanh thu sụt thì nới nhanh +1000bps/epoch lên lại trần 200%; không ai chỉnh tay được; (4) ngân sách mint chỉ tính stake−payout (payout đã trừ phí) một lần; mức launch 200% là subsidy có chủ đích, không hứa hedge farm luôn lỗ; (5) phí công khai 3.99% trên lệnh thắng: 0.99% về ví founder (owner thiết lập sau), 3.00% cho staker DIAMOND; founder chưa set thì phần đó dồn cho staker đang active; staker nhận theo stream 7 ngày, không ăn ngay bằng JIT stake; lệnh queue/trả một phần hoặc pool không đủ sau khi trả net thì không stream; chưa có staker thì phần staker ở lại pool; (6) DIAMOND khóa chuyển giao trong bootstrap, mở một chiều sau khi đủ thanh khoản, stake DIAMOND được chia doanh thu. Bộ kiểm thử xanh: 130 kiểm chứng hợp đồng, 51 kiểm chứng engine, 43 kiểm chứng security, fuzz 5 seed × ~160 bất biến. Rủi ro khai báo thẳng: trả chậm là có thật khi queue kẹt (claim vẫn được giữ bằng queue/credit nhưng có thể chờ; queueDebt không phải reserve ưu tiên của LP nên LP rút và winner chờ có thể cùng tồn tại), attester 2-trong-3 vẫn là điểm tin cậy (L1Read cross-check là lộ trình), và điều kiện pháp lý là cổng ra mắt.